

# Control is an illusion.

**Confidence** without control: scientific background and industry position.

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## ABSTRACT

Every result in cell biology is a difference from a control, and nothing checks the control against anyone else's. On the public JUMP Cell Painting dataset, a classifier names the institution from untreated wells alone **96.2%** of the time against 14.3% chance, so “normal” is a lab-specific coordinate, not a shared one. The standard batch corrections that make cross-lab discovery possible erase that coordinate (per-batch whitening drops lab identity to 12%, below chance): right for discovery, wrong for quality control. Datum Line keeps the coordinate and uses it. A lab's control-well images are placed against the published field and every lab running the same line, and the lab gets back a certificate, a distance, and the direction of any shift. The bet still to be proven is that label-free brightfield carries the same signal as five-stain fluorescence; the test is specified and pending.

## 1 The control problem

A control removes what a plate shares, including the plate's own state, so a sick plate hides behind its own controls. Write a measured well as a shared mean, a plate state, and a treatment effect that may depend on that state:

$$x_{test} = \mu + s + \tau(s) \quad x_{ctrl} = \mu + s \quad \Delta = x_{test} - x_{ctrl} = \tau(s) \quad (1)$$

The state  $s$  cancels from the number but not from the cells. If  $s$  is off,  $\Delta$  is still computed, still looks clean, and is measured on cells that are not the ones the field has in mind.

The same logic nests upward: well inside plate, plate inside batch, batch inside site, site inside the field. Each control sees only inside its own box. A whole plate off is visible to the batch; a whole site off is visible to nobody in the building. Only a reference outside every box, the field, can see where a lab sits.

This is not an exotic failure. Irreproducible preclinical research costs about \$28B a year in the US alone, and the largest single contributor, about 36%, is biological reagents and reference materials [1]. Cells are the input nobody certifies against anyone else's.

## 2 Morphological profiling as a sensor

An image of a cell is an unbiased, many-dimensional sensor: it answers “is anything about this run off?” rather than one hand-picked question. Cell Painting multiplexes six fluorescent dyes in five channels to mark eight cellular components, and automated analysis measures about 1,500 features per cell: size, shape, texture, intensity [2].

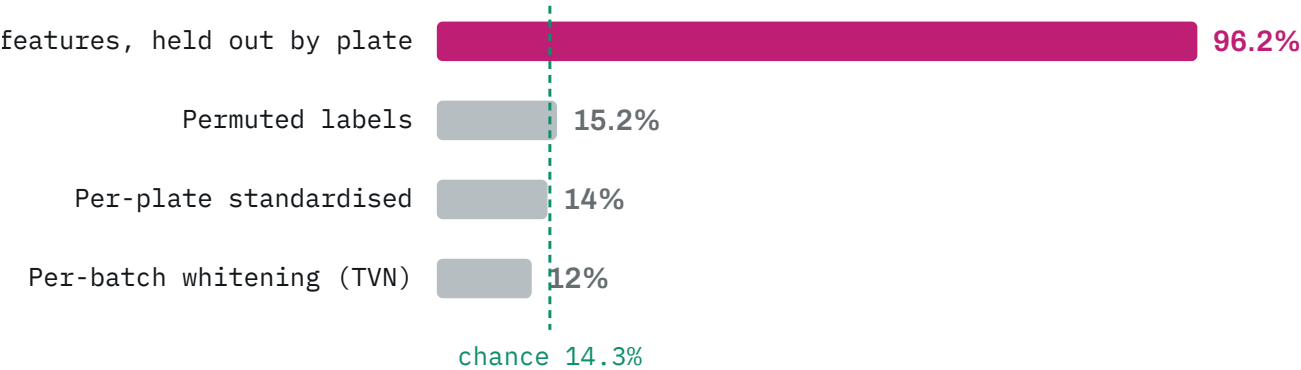
At scale, the reference already exists. The JUMP Cell Painting Consortium (ten pharmaceutical companies, six technology partners, two non-profits) profiled over 116,750 compounds plus gene knockouts and overexpressions, all in U2OS cells [3]. Its main compound set, cpg0016, was produced across 11 sites, which is exactly the variation a cross-lab reference needs, and it is public under CC0 [3].

The sensor is not the bottleneck. Many widely used bioactivity benchmarks are predicted well by simply counting cells, because their assays track cell health [5]. When a naive baseline matches a model, the comparison is the problem, not the model. In the author's hepatocyte work, three feature extractors (CellProfiler, a CNN, a ViT) performed similarly [6]. The asset is the reference, not the extractor.

## 3 Evidence on public data

Untreated wells carry their lab's fingerprint, and the normalizations built for discovery erase it.

Fig. 1 Lab identity from untreated wells, before and after correction

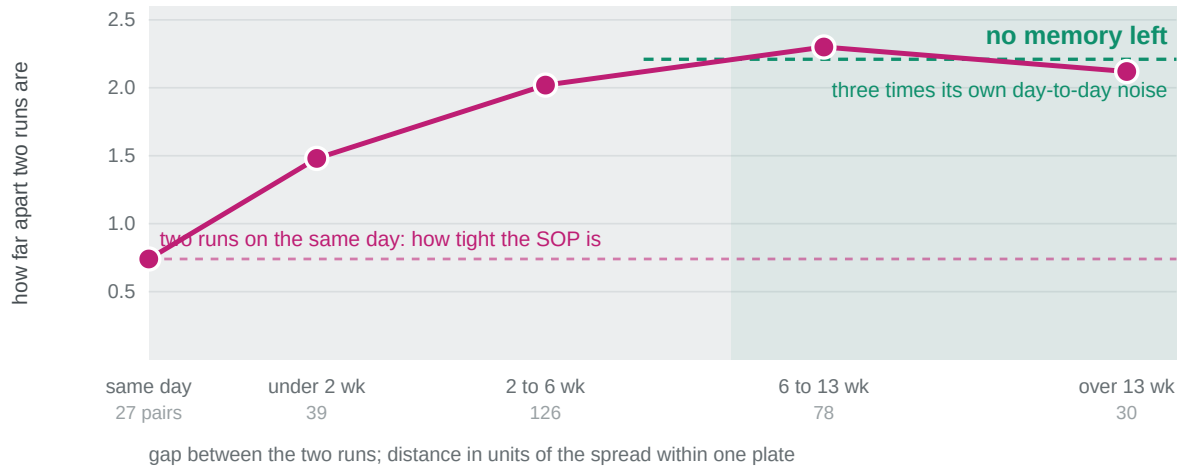


Balanced accuracy naming one of 7 institutions from vehicle-only JUMP wells; 12,046 wells, 3,781 features, folds held out by plate. results/e1.

| PLATE 1053600681                          | RESULT                                        | FILE       |
|-------------------------------------------|-----------------------------------------------|------------|
| Within-plate DMSO dispersion              | 79th percentile of 28; no local QC rule fires | results/e1 |
| Wells named as their labelled lab (lab 2) | 0 of 384                                      | results/e1 |

**Space.** The plate above passes every check its own lab can run, yet every one of its 384 vehicle wells sits in another institution’s territory. Local QC cannot see this by construction (section 1).

**Fig. 2 Even one lab drifts**



Distance between plate centroids of the same cells, in units of within-plate spread; one institution, 31 plates, 8 runs, June to October; same-day reference 0.74 (n = 27). results/e1b.

**Time.** Within one institution, the distance between two runs of the same cells grows from 0.74 for same-day pairs to about 3.1× that at 6 to 13 weeks, then stops: 2.30 then 2.12 in the last two bins. Drift is bounded, which is what makes a reference possible, and a lab’s own history stops being a valid comparison after about three months.

**Why correction is not the answer.** Batch correction for image-based profiling has been benchmarked on JUMP across ten methods [4]; its purpose is to remove lab and batch differences so biology can be compared. That is right for discovery. For quality control it deletes the one coordinate that says where a lab sits.

## 4 Formalism

Everything in section 3 is one of six operations on the same well vector, each a difference of means or a held-out classifier, so any lab can check it. Every well becomes an embedding, z-scored across the field:

$$e = \varphi(\text{image}) \in \mathbb{R}^{3781} \quad \varphi \in \{\text{CellProfiler}, \text{CNN}, \text{ViT}\} \quad (2)$$

| OPERATION                       | DEFINITION                                                         | ROLE                                               |
|---------------------------------|--------------------------------------------------------------------|----------------------------------------------------|
| 1 · Embedding                   | $e = \varphi(image)$                                               | the sensor; choice of $\varphi$ barely matters [6] |
| 2 · Lab classifier              | $f: e \rightarrow lab$                                             | does the lab leave a fingerprint (96.2%)           |
| 3 · Certificate distance        | eq. (3)                                                            | how far a plate sits from the field, and which way |
| 4 · Variance decomposition      | $Var(e) = V_{model} + V_{site} + V_{batch} + V_{plate} + V_{well}$ | how much is instrument, site, batch                |
| 5 · Residualize known variables | $e_{\perp} = e - \hat{E}[e   c]$                                   | remove only what is known to be instrument         |
| 6 · Morphology change           | $\delta = f(\ e - E[e   vehicle]\ )$                               | size of a shift, for the known-answer test         |

The certificate is a difference of two means in units of the field's spread, conditioned on the recorded variables  $c$  (cell line, plate type, microscope, and so on):

$$d = \| E[e | plate] - E[e | field, c] \|_{\Sigma} \quad (3)$$

Its direction names the features that moved; its negative points back toward the field. “Normal” is therefore a conditional expectation,  $E[x | every\ variable]$ , and it is only as good as the variables recorded. JUMP records the instrument side (site, batch, plate type, microscope, objective, filter set, pixel size); it does not publicly record serum or media lot, passage, seeding density, dye lots or operator.

**Discovery and QC want opposite things from the same vector.** Discovery wants invariance to the lab, so it aligns every plate to a reference before comparing. QC wants the lab coordinate, so it must never align it away.

## 5 Limits and open tests

Four claims are not yet proven, and each has a named test that could fail.

| OPEN QUESTION                                             | TEST                                                                                                 | WHAT WOULD FALSIFY IT                                        | STATUS   |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------|
| Does label-free brightfield carry the lab signal?         | Re-run e1 on the brightfield planes of the same 12,046 wells                                         | Brightfield near 14% chance: the certificate needs one stain | pending  |
| Does a lab's position change the answer for a fixed drug? | e8: 8 positive controls on 1,829 plates, 11 sources; effect vs the field's, against control distance | Spearman near zero: cells differ but answers do not          | pending  |
| Is the fingerprint biology or instrument?                 | Variance decomposition, microscope model separated from site                                         | Nearly all variance in $V_{model}$ : a calibration product   | pending  |
| Does drift generalize beyond one institution?             | Repeat e1b at each partner site                                                                      | Unbounded drift elsewhere: no stable reference               | one site |

Already controlled: vehicle-only wells (no compounds, so no treatment signal to leak), folds held out by plate, a permuted-label baseline at chance (15.2%), and the result surviving removal of the 1536-well source.

## 6 Industry position

The market checks cell identity and ignores cell behaviour. Datum Line is the behaviour half, held by nobody who sells cells, media or instruments.

| OFFERING TODAY                      | WHAT IT PROVES           | WHAT IT CANNOT SEE                    | RELATION                             |
|-------------------------------------|--------------------------|---------------------------------------|--------------------------------------|
| STR authentication                  | which line it is         | same fingerprint, different behaviour | complement: identity, then behaviour |
| Vendor certificate of analysis      | the lot shipped to spec  | your incubator, after passage         | complement                           |
| Instrument software, plate controls | the plate against itself | a whole plate, batch or site off      | reads the same images                |

| OFFERING TODAY                        | WHAT IT PROVES                  | WHAT IT CANNOT SEE                | RELATION                         |
|---------------------------------------|---------------------------------|-----------------------------------|----------------------------------|
| Phenomics platforms, batch correction | biology, lab removed on purpose | the lab coordinate, by design     | collaborator: discovery upstream |
| Eyeballing and cell counts            | gross death or overgrowth       | texture and shape at equal counts | what labs do now                 |

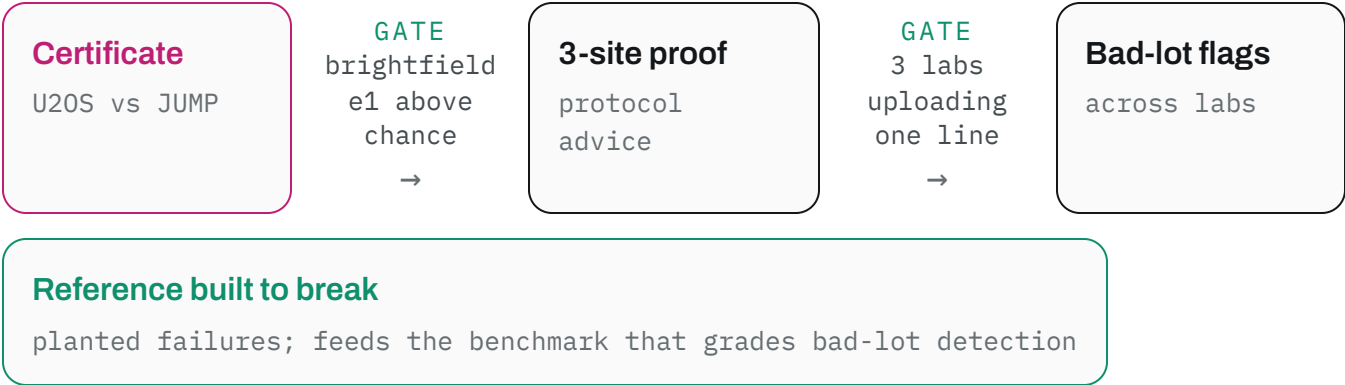
**Who pays.** Academic labs and cores, on every new line (at stake: the paper). Pharma and CRO screening groups, on every transfer (at stake: the screen; one line checked late cost about \$2.6M in the author’s experience: 3 runs a week for 17 weeks at about \$50k a run). Cell banks and vendors, on every lot (at stake: the lot and every lab that bought it).

**Why it compounds.** Day one needs no network: the public JUMP reference already holds seven institutions’ U2OS controls, so the first customer is compared with the field, not with an empty database. Every upload widens the reference, and bad-lot detection needs many labs on the same lot, so the later products exist only once the network does.

**Price.** Per line, per quarter, set by the drift plateau: one reading covers its whole useful life. The figure is not yet set. **What leaves the building:** untreated control-well images only; no compounds, targets or results.

## 7 Roadmap

**Fig. 3** Each phase opens only when the test before it passes



Gates are falsifiable tests from section 5; size and budget of the reference set are not yet set.

## References

References 1 to 5 were opened and checked for this paper. Reference 6 is the author's own work. Results files (results/e1, e1b, e8) are the author's analyses of cpg0016.

1. Freedman LP, Cockburn IM, Simcoe TS. The economics of reproducibility in preclinical research. *PLoS Biology* 13(6):e1002165, 2015. [PMC4461318](#)
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5. Seal S, Dee W, ... Titterton K, ... Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nature Communications*, 2026. [doi:10.1038/s41467-026-68725-5](#)
6. Ewald JD, Titterton KL, et al. Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes. *Cell Systems*, 2026.